

CFD Analysis of a Converging–Diverging Rocket Nozzle

1. Project Overview

This project investigates compressible flow through a 2D axisymmetric converging–diverging rocket nozzle using ANSYS Fluent. The objective was to study supersonic flow behavior, Compressible Expansion Behaviour, and the effect of varying outlet pressure conditions on nozzle performance.

The simulation focused on:

- Compressible flow Visualization
 - Compressible flow modelling
 - Structured mesh generation
 - Pressure ratio effects
 - Flow parameter extraction
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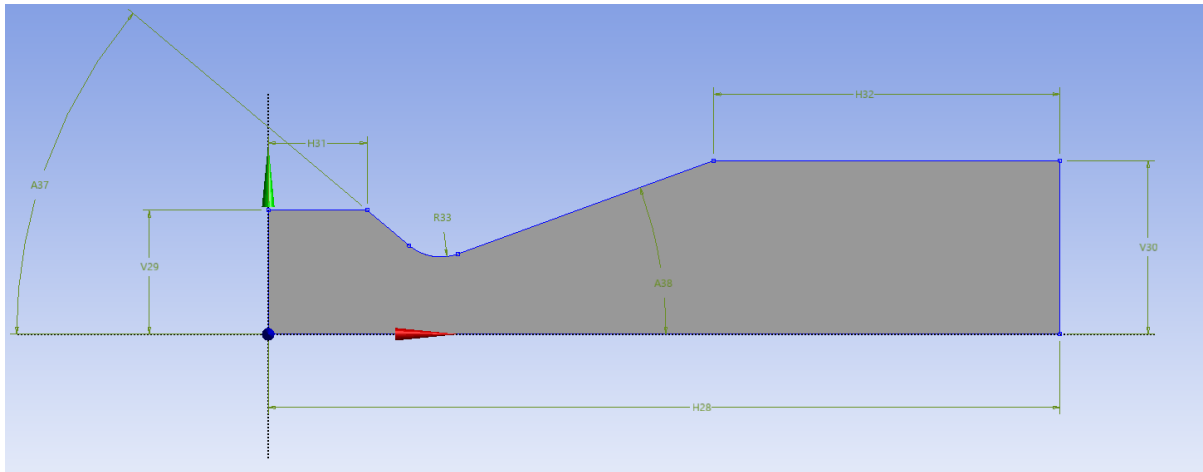
2. Objectives

The main objectives of this project were:

- Create a 2D axisymmetric nozzle geometry
 - Generate a high-quality structured quadrilateral mesh
 - Simulate compressible flow using a density-based solver
 - Analyze Mach number and pressure distributions
 - Observe over-expanded nozzle behavior
 - Visualize shock structures using contour plots
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3. Geometry Description

A converging–diverging nozzle geometry was created in ANSYS Design Modeler



Assumptions

- 2D axisymmetric domain
- Steady compressible flow
- Ideal gas behavior
- Adiabatic nozzle walls

Geometry Parameters

Parameter	Value
Nozzle Length	320 mm
Throat Diameter	62.308 mm
Exit Diameter	140 mm
Area Ratio	5.048

4. Mesh Generation

A structured quadrilateral mesh was generated to accurately capture shock waves and minimize numerical diffusion.

Mesh Strategy

- Face-mapped quadrilateral mesh
- Local refinement near throat region
- Boundary layer refinement near walls

- Increased cell density near expected shock regions

Mesh Quality

Metric	Value
Total Cells	32400
Minimum Orthogonal Quality	0.5608
Maximum Aspect Ration	8

Figure 2: Structured quadrilateral mesh of the convergent-divergent nozzle consisting of 32400 cells

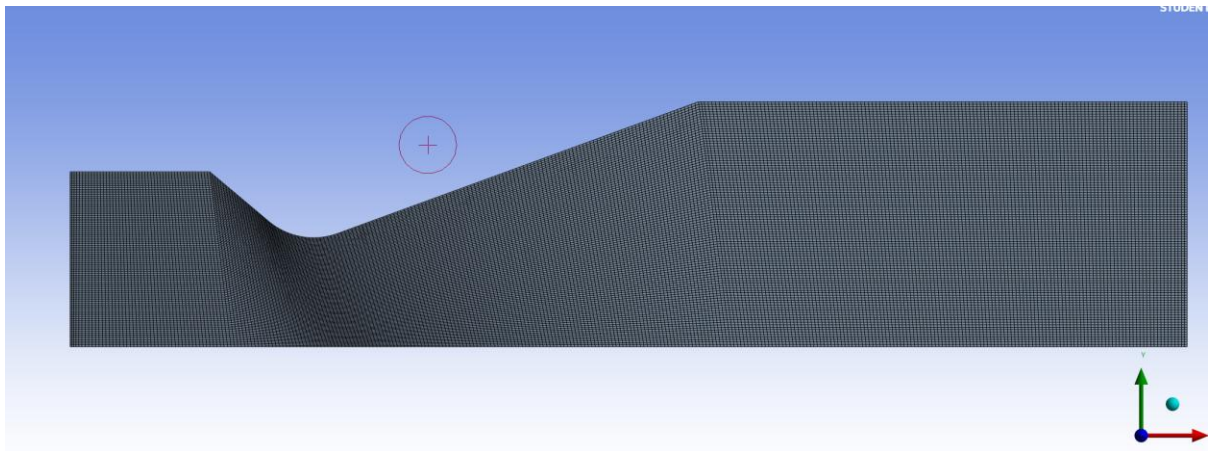
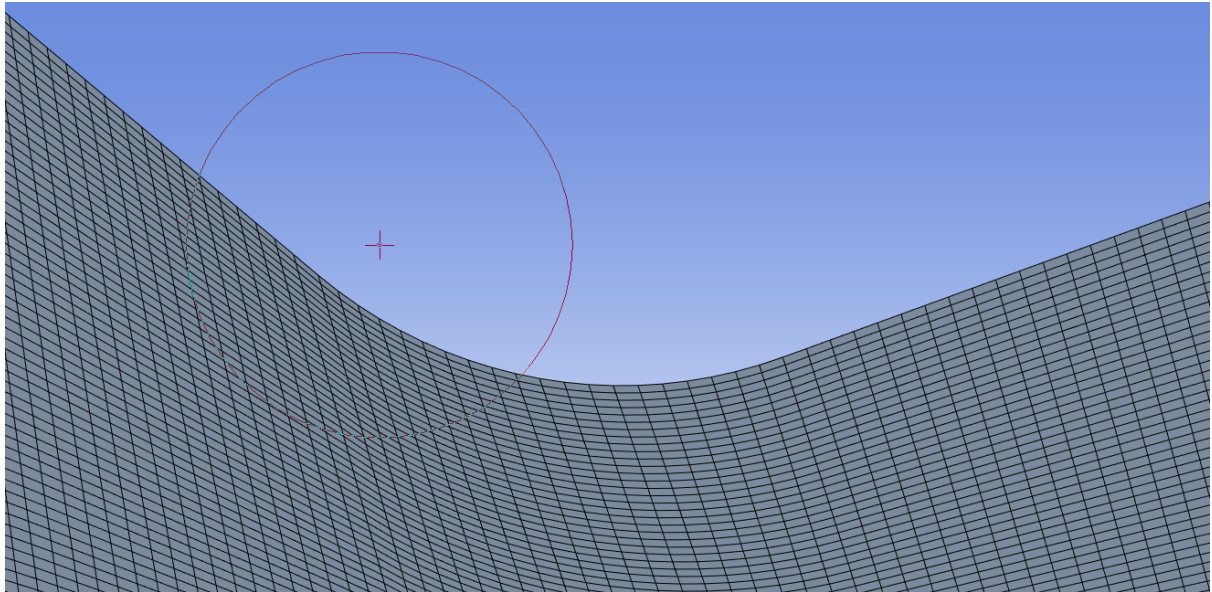


Figure 3: Detailed view of the throat region showing structured mesh alignment and local resolution near the minimum-area section.



5. Solver Setup

The simulation was performed using ANSYS Fluent.

Solver Configuration

Setting	Value
Solver Type	Density-Based
Time Formulation	Steady
Turbulence Model	Inviscid
Energy Equation	Enabled
Fluid	Air (Ideal Gas)

Boundary Conditions

Boundary	Condition
Inlet	Pressure Inlet
Inlet Total Pressure	100000 Pa
Inlet Total Temperature	300k
Outlet	Pressure Outlet
Outlet Pressure	10000 Pa

6. Case Studied

The outlet pressure condition investigated.

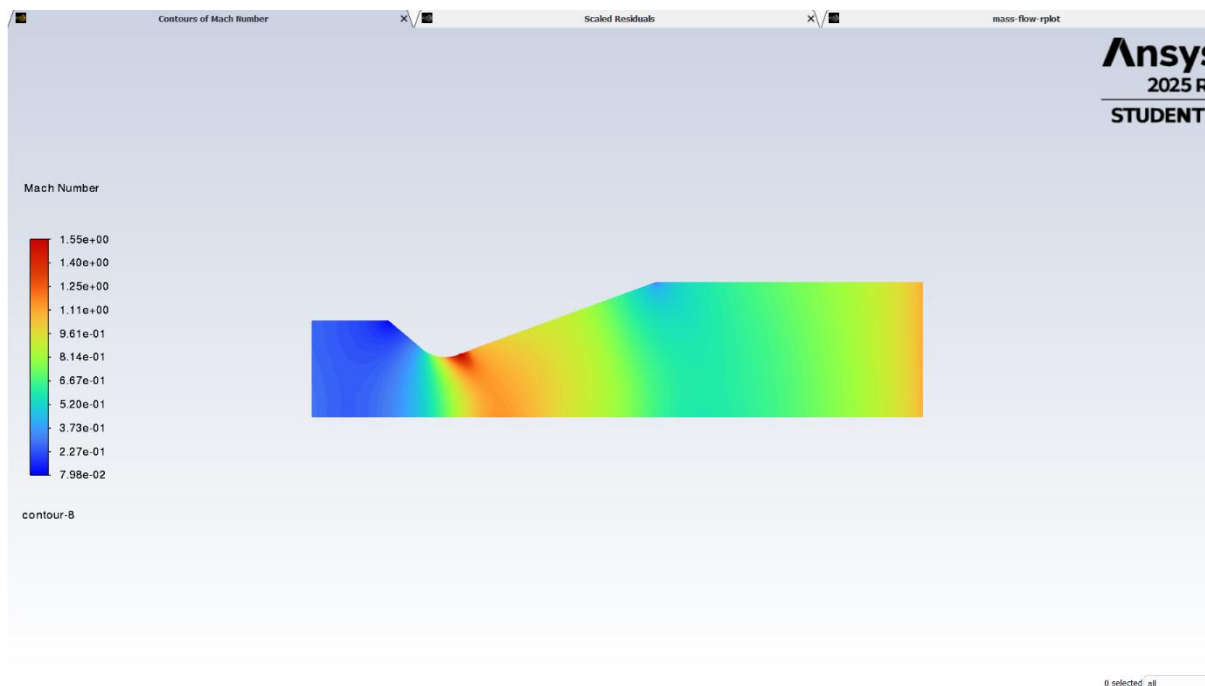
Case Outlet Pressure Flow Condition

Case A 10000 Pa Over-expanded

7. Results and Discussion

7.1 Mach Number Contours

The Mach number reached approximately 1.5 near the throat region, followed by a non-uniform downstream distribution. Since no sharp pressure rise was visible in the pressure contour, the downstream Mach variation was treated as numerical/flow-field instability rather than a clearly resolved shock.

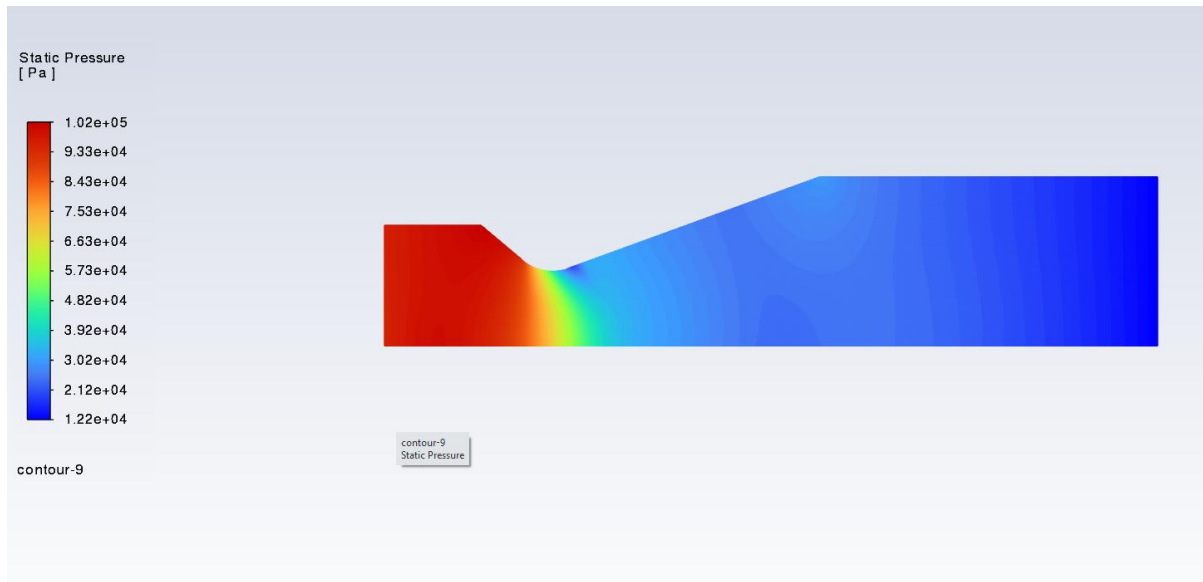


Observations

- Mach number reached approximately 1.5
 - Supersonic expansion observed downstream of throat
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7.2 Pressure Contours

The static pressure contour shows a continuous pressure drop from the inlet toward the nozzle exit. No clearly defined normal shock was observed in this case. The result indicates expansion of the compressible flow through the diverging section, although convergence oscillations limited the quantitative accuracy of the downstream solution.

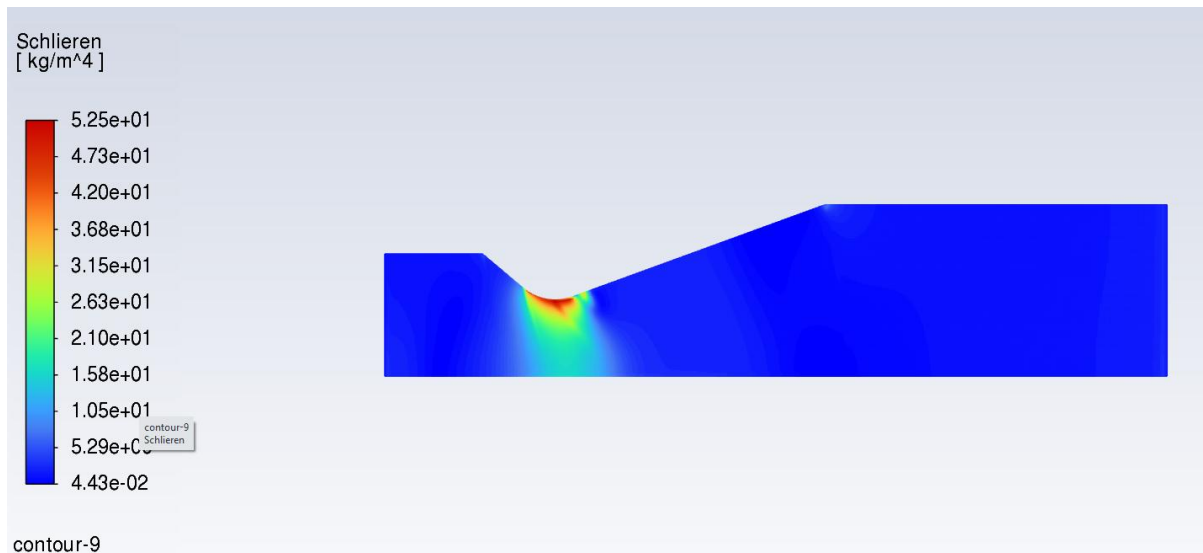


Observations

- Maximum pressure located near inlet
- Pressure drops observed after throat
- Continuous pressure decreases through the Nozzle
- No clear pressure Jump indicating an internal shock

7.3 Schlieren / Density Gradient Visualization

The Schlieren visualization highlights regions of strong density variation within the nozzle. The highest density gradients occurred near the throat, where the flow accelerated rapidly from subsonic toward supersonic conditions. Downstream of the throat, the density gradients gradually weakened, indicating continued expansion of the flow. No distinct shock wave was identified in the diverging section, which is consistent with the smooth pressure distribution observed in the static pressure contour.



Observations

- Maximum density gradients were observed near the nozzle throat.
- Strong acceleration of the flow produced rapid density variations in this region.
- The density gradient decreased downstream of the throat as the flow expanded through the diverging section.
- No clearly defined shock structure was observed within the computational domain.

7.4 Centerline Data Extraction

Flow variables were examined qualitatively using contour plots throughout the nozzle domain. Due to convergence oscillations and the focus of this project on flow visualization, detailed centreline data extraction was not included. Future work may include extraction of Mach number, pressure, and temperature distributions along the nozzle centreline for quantitative comparison with analytical compressible flow theory.

8. Validation

The CFD results were compared qualitatively with classical compressible flow theory. The flow accelerated from subsonic conditions at the inlet toward Mach 1 near the throat and further accelerated in the diverging section. The static pressure continuously decreased through the nozzle, while the Schlieren visualization confirmed strong density gradients near the throat. The observed trends agree with the expected behaviour of a converging-diverging nozzle.

Observed behavior:

- Mach ≈ 1 at throat.

- Maximum Mach ≈ 1.5 .
 - Pressure decreases from ≈ 102 kPa to ≈ 12 kPa.
 - Strong density gradients near throat.
 - No clearly resolved internal shock.
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9. Challenges Encountered

- Initial convergence instability and oscillations
- CFL sensitivity during startup
- Mesh refinement near throat region
- Issue with mesh Refinement
- Difficulty reducing the mass imbalance; Lowest achieved mass imbalance was approximately 6.7%
- Had to re-initialise with different Values to achieve a good result.

These issues were improved through structured meshing and solver tuning.

10. Conclusion

The project successfully demonstrated compressible flow behavior inside a converging-diverging rocket nozzle using ANSYS Fluent. The simulation captured the acceleration of flow from subsonic to supersonic conditions and illustrated the corresponding pressure and density changes. While convergence oscillations limited the quantitative accuracy of the final solution, the simulation reproduced the key physical characteristics predicted by compressible flow theory.

Key outcomes:

- Successful axisymmetric CFD model.
- High-quality structured mesh generation.
- Supersonic flow acceleration.
- Pressure and density field visualization.
- Experience with density-based solvers and CFD troubleshooting.

The project provided practical experience in mesh generation, solver configuration, convergence assessment, and interpretation of compressible flow results.

11. Future Improvements

Possible future extensions include:

- Mesh independence study
 - Comparison with analytical nozzle theory
 - Thrust calculation
 - Improve convergence stability and reduce mass imbalance through further mesh optimization and solver tuning.
 - Adaptive mesh refinement
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